

Tutorial No.8

Q.I] Justify whether the following Linear Transformation A, is Regular, Singular, or Orthogonal, Justify your answer

i) $A = \frac{1}{13} \begin{bmatrix} -12 & -5 \\ 5 & -12 \end{bmatrix}$

ii) $A = \frac{1}{15} \begin{bmatrix} 5 & -14 & 2 \\ -10 & -5 & -10 \\ 10 & 2 & -11 \end{bmatrix}$

Q.II If $T: R^4 \rightarrow R^3$ is defined as,

$T(x_1, x_2, x_3, x_4) = (x_1 - x_2 + 2x_3 + x_4, x_1 - x_2 + 3x_3 + 2x_4, 2x_1 - 2x_2 + 4x_3 + 2x_4)$
then find basis for range and kernel of T. Hence determine rank and nullity of T and verify the rank nullity theorem.

Q.III] Find the matrix of the Linear Transformation $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ which produces the effect of shear of factor 3 along positive Y direction, followed by reflection about $y = -x$, followed by clockwise rotation through an angle of 30° .

Hence find the image of $u = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ under the transformation T.